

BC660K-GL

Reference Design

NB-IoT Module Series

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Status: Released



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About the Document

Revision History

Version	Date	Author	Description
-	2020-09-30	Clifton HE	Creation of the document
1.0	2021-01-20	Clifton HE	First official release
1.1	2021-04-25	Clifton HE	Sheet 6: Modified the notes about the UART Interface; Added a note about the application of the level conversion circuit.

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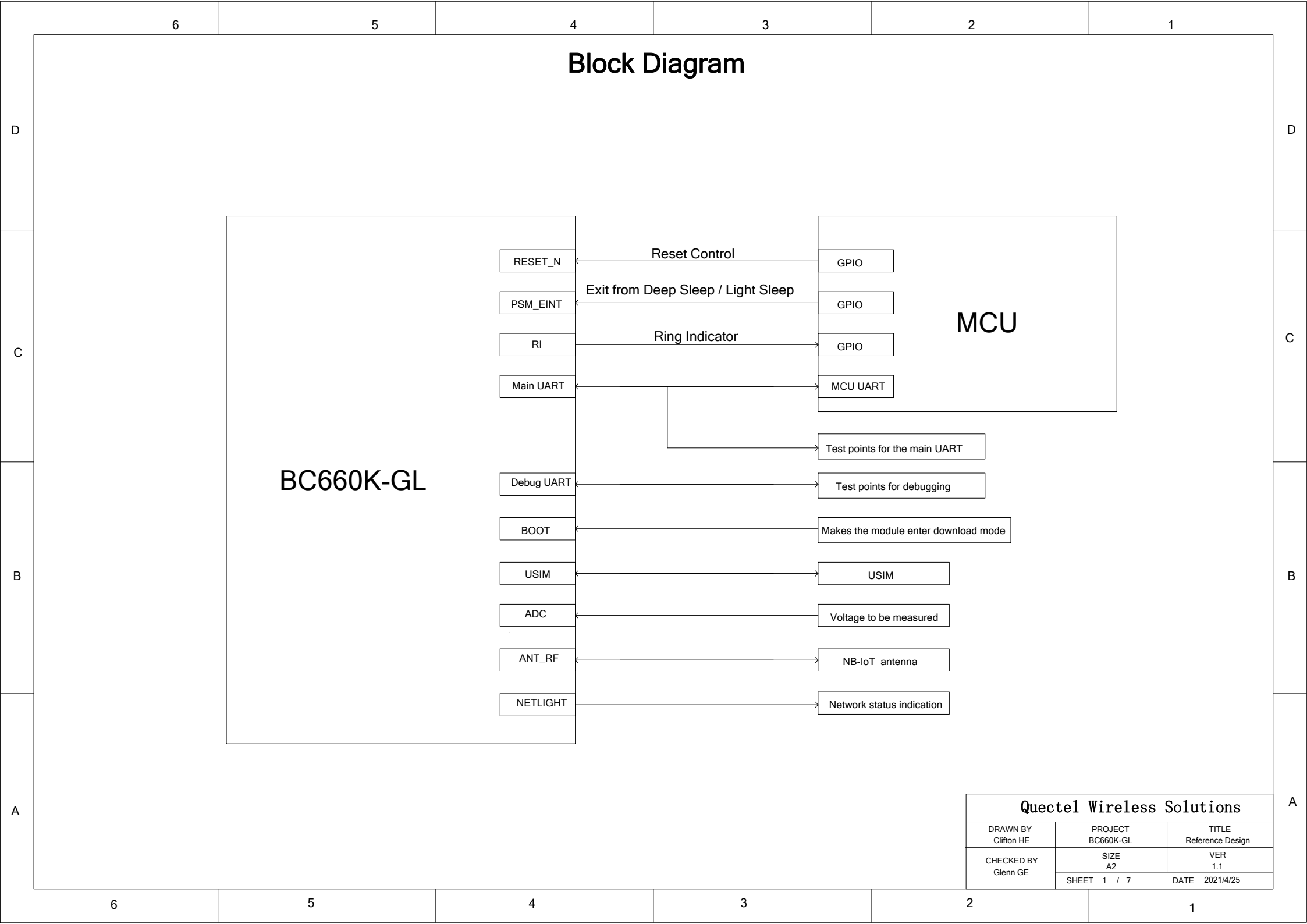
1 Reference Design

1.1. Introduction

This document provides the reference design for Quectel BC660K-GL module, including the design for power supply, module interfaces, etc.

1.2. Schematics

The schematics illustrated in the following pages are provided for reference only.



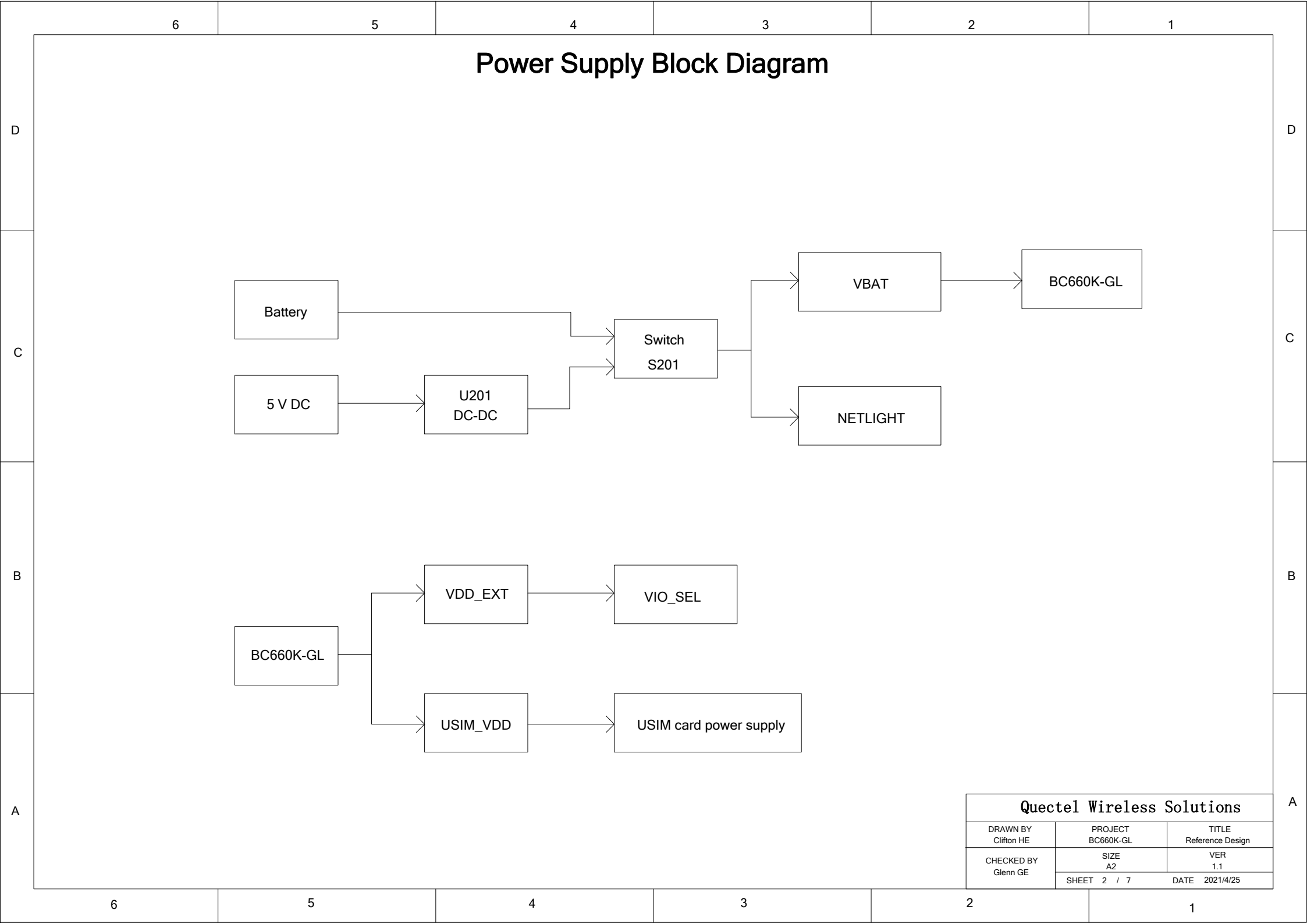
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Power Supply Block Diagram

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graph LR; Battery[Battery] --> S201[Switch S201]; 5VDC[5 V DC] --> U201[U201 DC-DC]; U201 --> S201; S201 --> VBAT[VBAT]; S201 --> NETLIGHT[NETLIGHT]; VBAT --> BC660K_GL1[BC660K-GL]; BC660K_GL2[BC660K-GL] --> VDD_EXT[VDD_EXT]; BC660K_GL2 --> USIM_VDD[USIM_VDD]; VDD_EXT --> VIO_SEL[VIO_SEL]; USIM_VDD --> USIM_supply[USIM card power supply];
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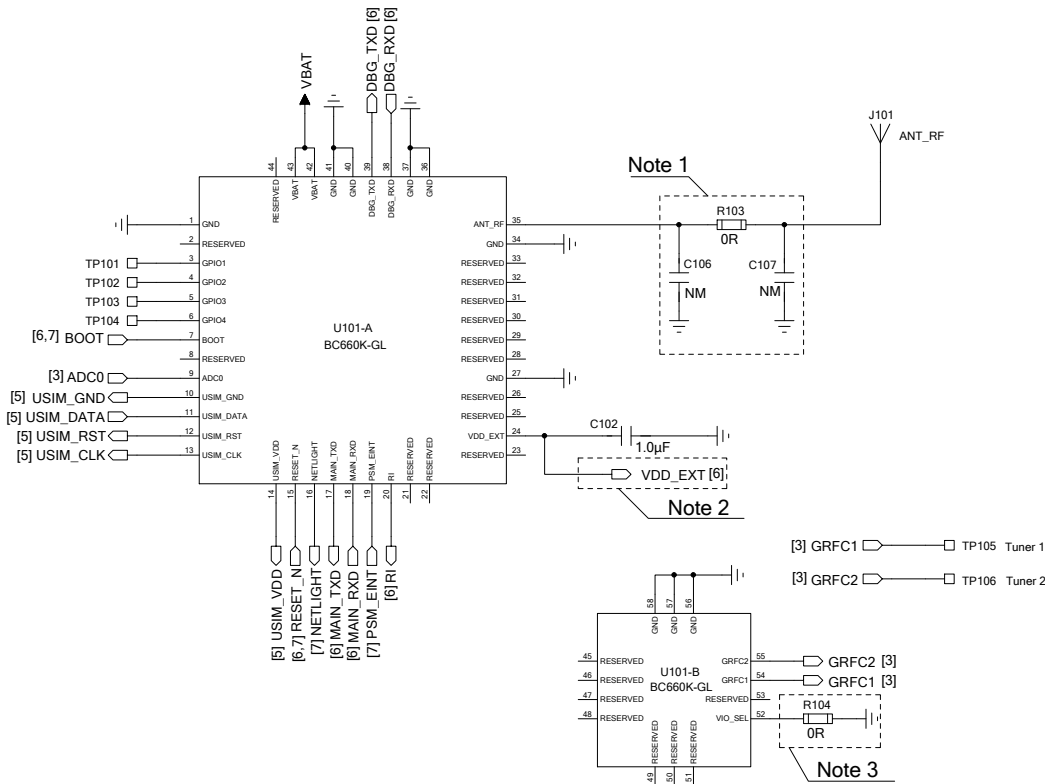
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D	Power Supply Block Diagram						D												
C	<pre>graph LR; Battery --> S201; 5V_DC[5 V DC] --> U201[U201 DC-DC]; U201 --> S201; S201 --> VBAT; S201 --> NETLIGHT; VBAT --> BC660K_GL[BC660K-GL]; BC660K_GL --> VDD_EXT[VDD_EXT]; BC660K_GL --> USIM_VDD[USIM_VDD]; VDD_EXT --> VIO_SEL[VIO_SEL]; USIM_VDD --> USIM_card[USIM card power supply];</pre>						C												
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A	<table><tr><td colspan="3">Quectel Wireless Solutions</td></tr><tr><td>DRAWN BY Clifton HE</td><td>PROJECT BC660K-GL</td><td>TITLE Reference Design</td></tr><tr><td>CHECKED BY Glenn GE</td><td>SIZE A2</td><td>VER 1.1</td></tr><tr><td colspan="2">SHEET 2 / 7</td><td>DATE 2021/4/25</td></tr></table>						Quectel Wireless Solutions			DRAWN BY Clifton HE	PROJECT BC660K-GL	TITLE Reference Design	CHECKED BY Glenn GE	SIZE A2	VER 1.1	SHEET 2 / 7		DATE 2021/4/25	A
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Module Interfaces

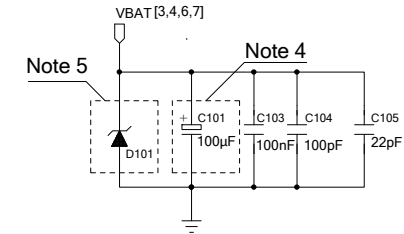
Module Interfaces



Notes:

1. A n-type matching circuit is recommended in RF layout.
For more details, see *Quectel_RF_Layout_Application_Note*.
2. VDD_EXT is a 1.8/3.3 V output power supply and has no voltage output in Deep Sleep/Light Sleep mode. It is used to supply power for the module's pull-up circuits, and thus is not recommended to be used as the power supply for external circuits.
3. The VIO_SEL is used to control VDD_EXT voltage selection.
When VIO_SEL is floating (R104 not mounted), the VDD_EXT voltage domain is 1.8 V;
when VIO_SEL is grounded (R104 mounted), and VBAT ≥ 3.3 V, the VDD_EXT voltage domain is 3.3 V;
when VIO_SEL is grounded (R104 mounted), and VBAT < 3.3 V, the VDD_EXT voltage domain equals to VBAT voltage.

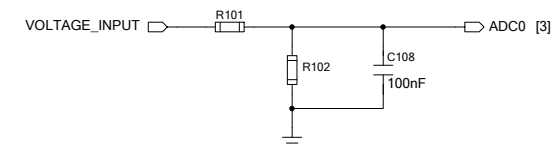
VBAT Input



Notes:

1. The input voltage of VBAT ranges from 2.2 V to 4.3 V.
2. The width of VBAT trace is recommended to be wider than 0.5 mm, and the longer the trace is, the wider it should be.
3. The capacitors should be placed in ascending order of the capacitance value, and the one with the minimum capacitance should be placed nearest to the VBAT pins. Additionally, all these capacitors should be placed as close to the VBAT pins as possible.
4. The capacitance of C101 should be chosen by debugging to ensure that the voltage of VBAT will not fall below the lowest operation voltage of the module during bursting.
5. It is recommended to add a TVS close to the VBAT pin.

ADC



Notes:

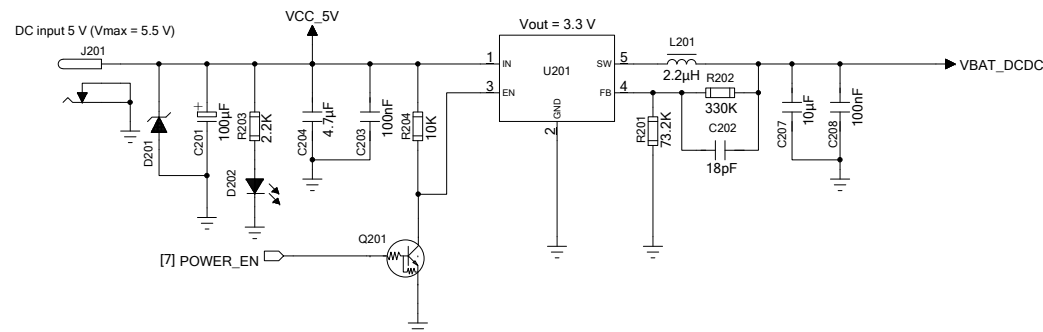
1. The maximum output voltage of the 12-bit ADC is 1.2 V.
2. A 320 kΩ pull-down resistor is integrated inside the ADC pin.
This resistor needs to be considered when you calculate the voltage division relationship.

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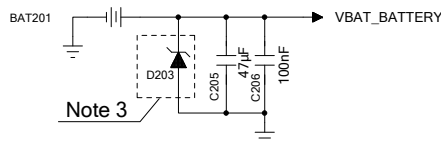
Power Supply

DC Power Supply



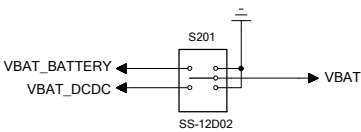
Note:
The output current of the power converter should be no less than 0.5 A.

Battery Application



- Notes:
1. The battery voltage shall range between 2.2 V and 4.3 V to meet the module's power supply requirements, and the battery's rated output current should be higher than the module's maximum current consumption. Additionally, it is recommended to do reverse battery protection to avoid damages to the module.
 2. For different batteries selected, the capacitance of C205 should be adjusted appropriately according to the debugging results.
 3. It is recommended to add a TVS close to the battery.

Power Supply Selection

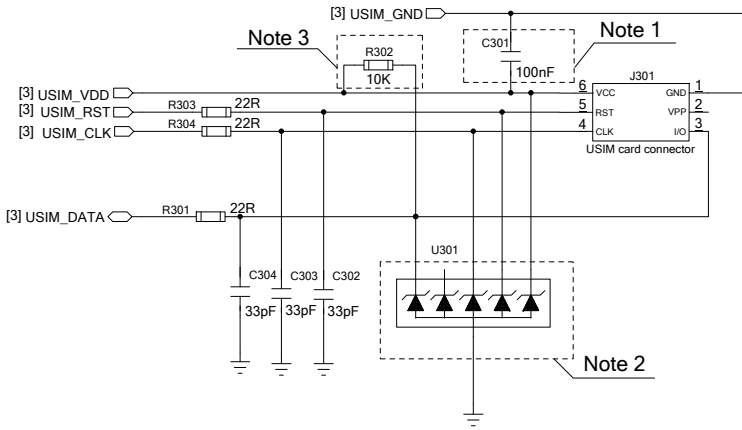


- Notes:
1. S201 is used to switch between an external 5 V power supply and a battery power supply.
 2. VBAT ranges from 2.2 V to 4.3 V, and the typical value is 3.3 V.

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USIM Interface

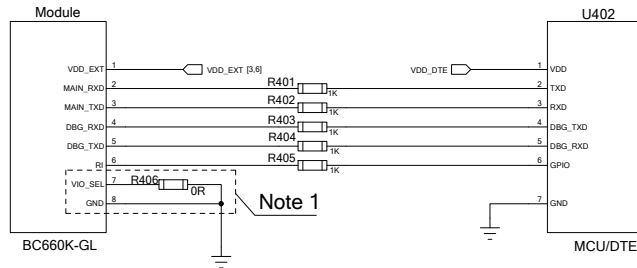


- Notes:
1. The value of C301 should be less than 1 μ F.
 2. U301 is used for protecting the USIM interface against ESD, and the junction capacitance should be less than 50 pF. It should be placed nearby the USIM card connector.
 3. It is necessary to add a 10 k Ω pull-up resistor between USIM_DATA and USIM_VDD to improve anti-interference ability.
 4. For more design guidelines, see *Quectel_BC660K-GL_Hardware_Design*.

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UART Interface

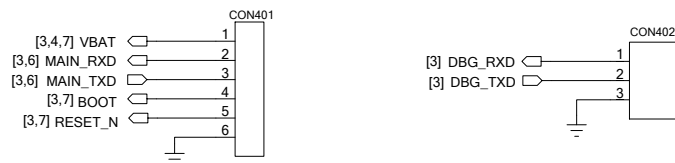
Reference Design for UART



Notes:

- When VIO_SEL is grounded and VBAT < 3.3 V, VDD_EXT = VBAT;
When VIO_SEL is grounded and VBAT ≥ 3.3 V, VDD_EXT = 3.3 V;
When VIO_SEL is floating, VDD_EXT = 1.8 V.
If the voltage domain of your application system is 1.8 V, do not mount R406;
If the voltage domain of your application system is 3.3 V, mount R406;
- When a URC or message is received, the module will inform the DTE through the RI pin.
- MAIN_RXD cannot be pulled-up to VDD_EXT directly.
- Pay attention to the level matching issue of UART ports during application.

Recommended Test Points for Firmware Upgrade and Debugging

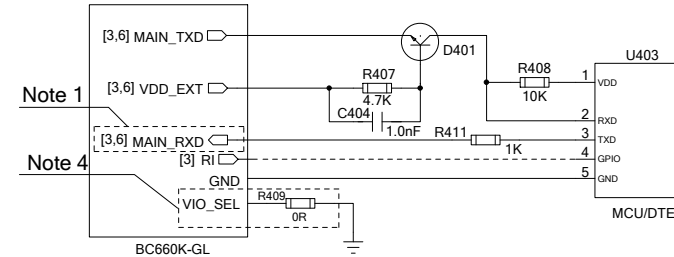


Note:

Pay attention to the level matching issue of the port during application.

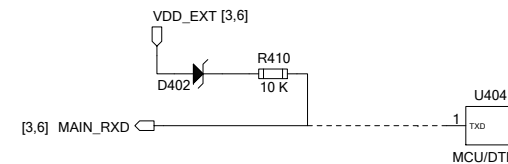
UART Level Translation - Level Conversion Circuit Solution

When the voltage domain of your application system is neither 1.8 V nor 3.3 V, it is recommended to use a level conversion circuit.



Notes:

- MAIN_RXD cannot be pulled up to VDD_EXT directly. Due to the anti-backflow design of the MAIN_RXD pin, it can be directly connected to MCU_TXD in 1.8-3.3 V voltage domain.
If the Deep Sleep/Light Sleep mode wake-up function of the MAIN_RXD pin is enabled, it is not recommended to use a level conversion circuit in MAIN_RXD to avoid abnormal wake-up.
- When a URC or message is received, the module will inform the DTE through the RI pin.
- Pay attention to the level matching issue of UART ports during application.
- While using the level conversion circuit, do not mount R409.
- The level conversion circuit does not apply to applications with high baud rates exceeding 460 kbps.
- For the design of the circuit shown in the dotted line (RI-GPIO circuit), see that of the MAIN_TXD-RXD circuit, please note the connecting direction at the same time.



Notes:

- When MCU_TXD is greater than 3.3 V, MAIN_RXD cannot be directly connected to MCU_TXD, it is required to use level conversion.
- To pull up MAIN_RXD to VDD_EXT, you need to connect a Schottky diode in series first, and then pull up MAIN_RXD through a 4.7-20 kΩ resistor.

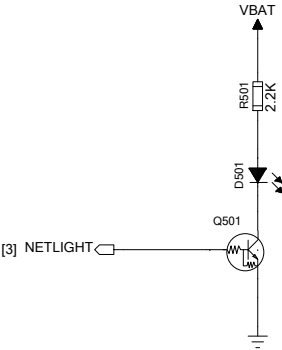
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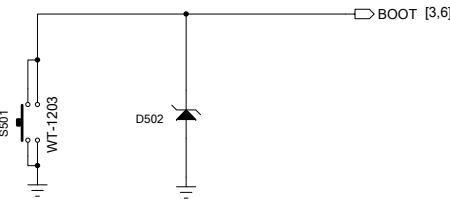
MCU Control and Drive

Network Status Indication

The NETLIGHT pin is used to indicate network status.

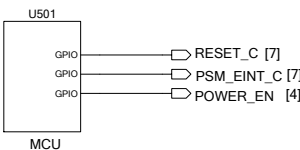


BOOT



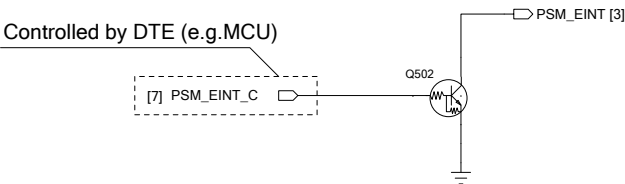
Note:
If the BOOT pin is connected to a filter capacitor in parallel, the capacitance of the capacitor cannot be higher than 33 pF.

MCU GPIOs

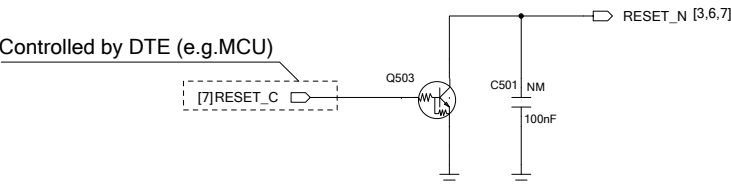


PSM_EINT

PSM_EINT can be used to wake up the module from Deep Sleep/Light Sleep.

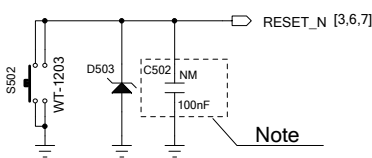


RESET_N
MCU Application



Note:
It is recommended to reserve a 100 nF capacitor close to the RESET_N pin.

RESET_N
Keystroke Application



Note:
It is recommended to reserve a 100 nF capacitor close to the RESET_N pin.

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